Navigating AI in evidence synthesis

AI tools across the review workflow

Luke Baxter

Map and presentation

lukebax.github.io/ai-review-tools/

Stages of a systematic review

1 Protocol

2 Search

3 Screening for relevance

4 Quality assessment

5 Data extraction

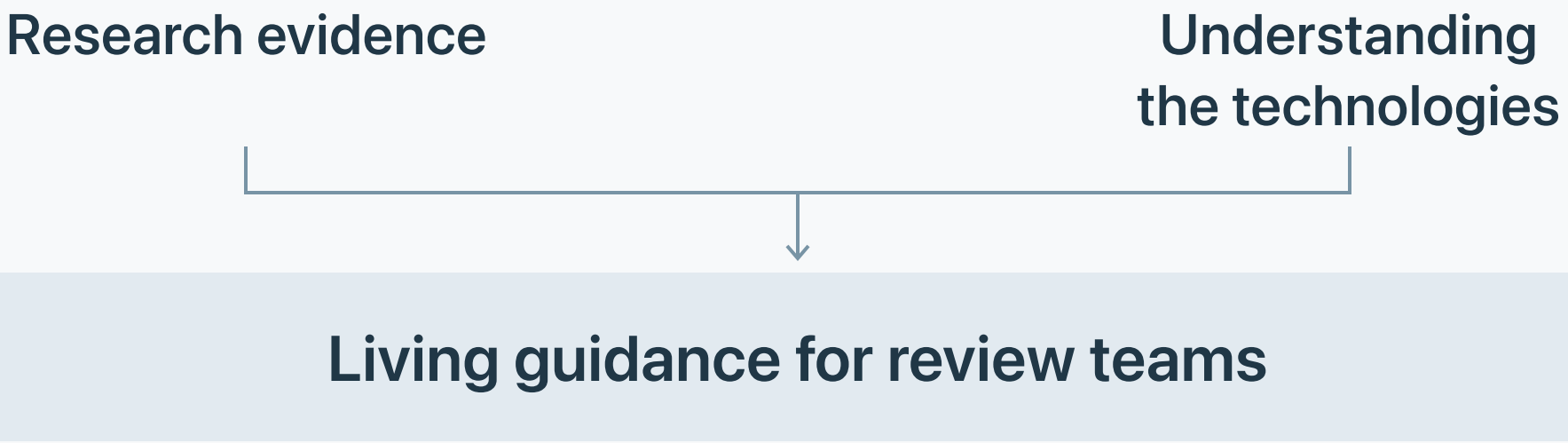
6 Data synthesis

7 Certainty of evidence

8 Manuscript

RAISE provides guidance for using AI in reviews

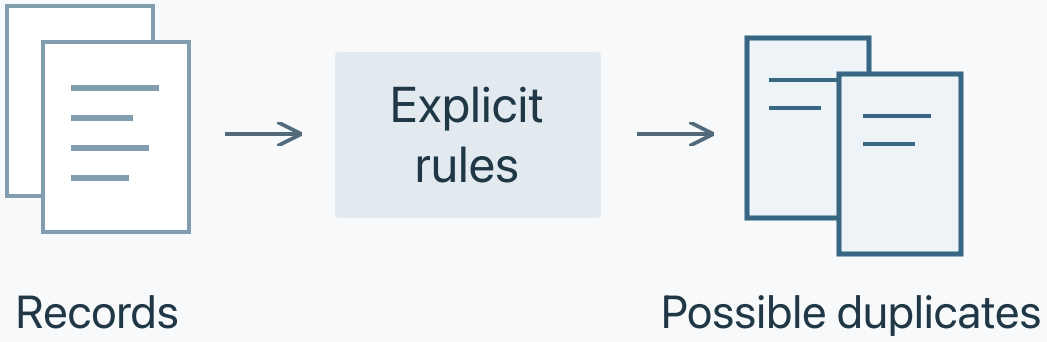
Responsible use of AI in evidence synthesis



Developed by researchers working in evidence synthesis

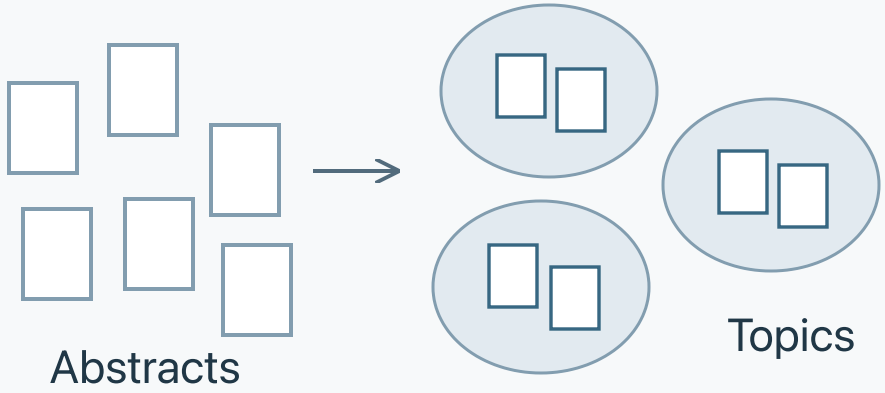
Explicit rules and patterns in unlabelled data

Rule-based algorithms



People define the rules used to compare records.

Unsupervised machine learning

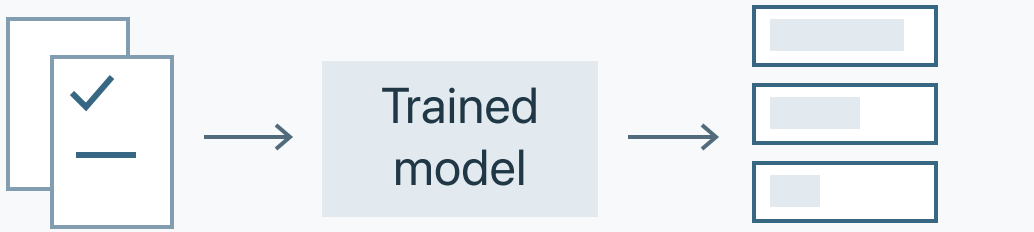


Patterns emerge without supplied labels.

Results depend on the records and the method.

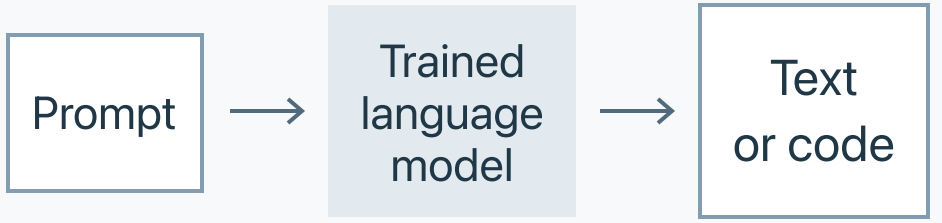
Learning from labelled examples and generating language

Supervised machine learning



Already trained or trained during the review

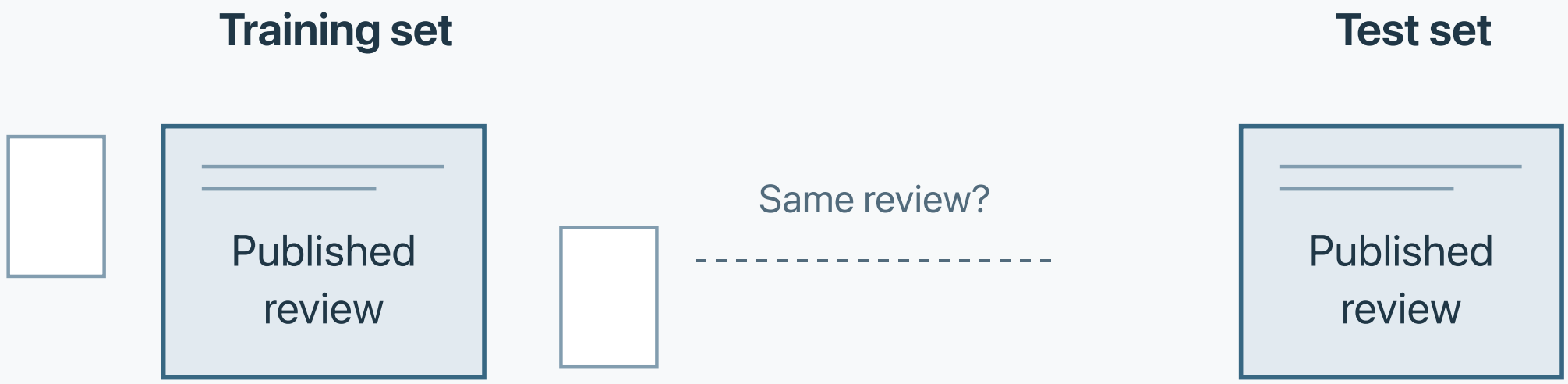
Generative LLMs



Outputs can vary.
Fluent output can be wrong.

Evaluation needs independence from development

Possible training–test overlap: a hypothetical example



Apparent performance may overstate performance on new material.

Incomplete disclosure makes overlap harder to establish.

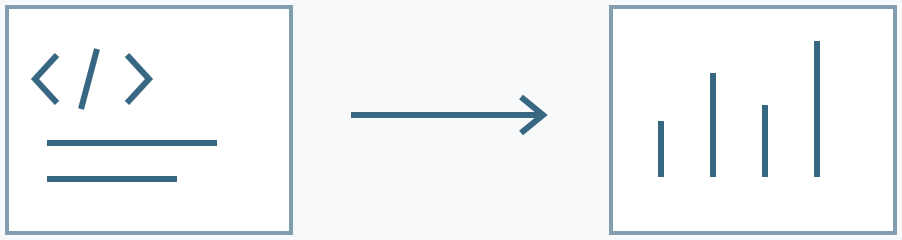
Recommendations apply to a particular use

Acceptable for use	Use outputs, accounting for limitations and bias
Human verification required	Check outputs and correct errors
Requires validation within the review	Test performance against a prespecified threshold and suitable benchmark
Exploratory and supplementary use	Explore or supplement established methods
Not acceptable for use	Do not rely on outputs for this task

Verification and validation ask different questions

Verification

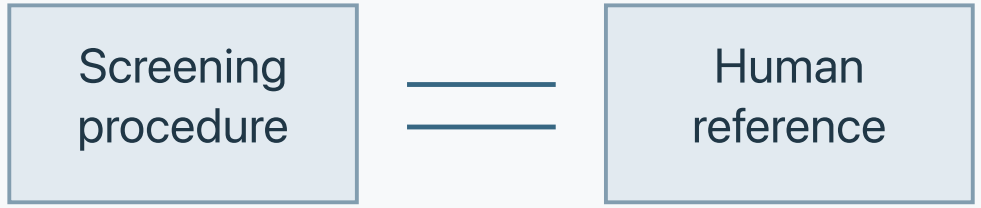
Check the output



Does it carry out the intended analysis?

Validation

Evaluate performance



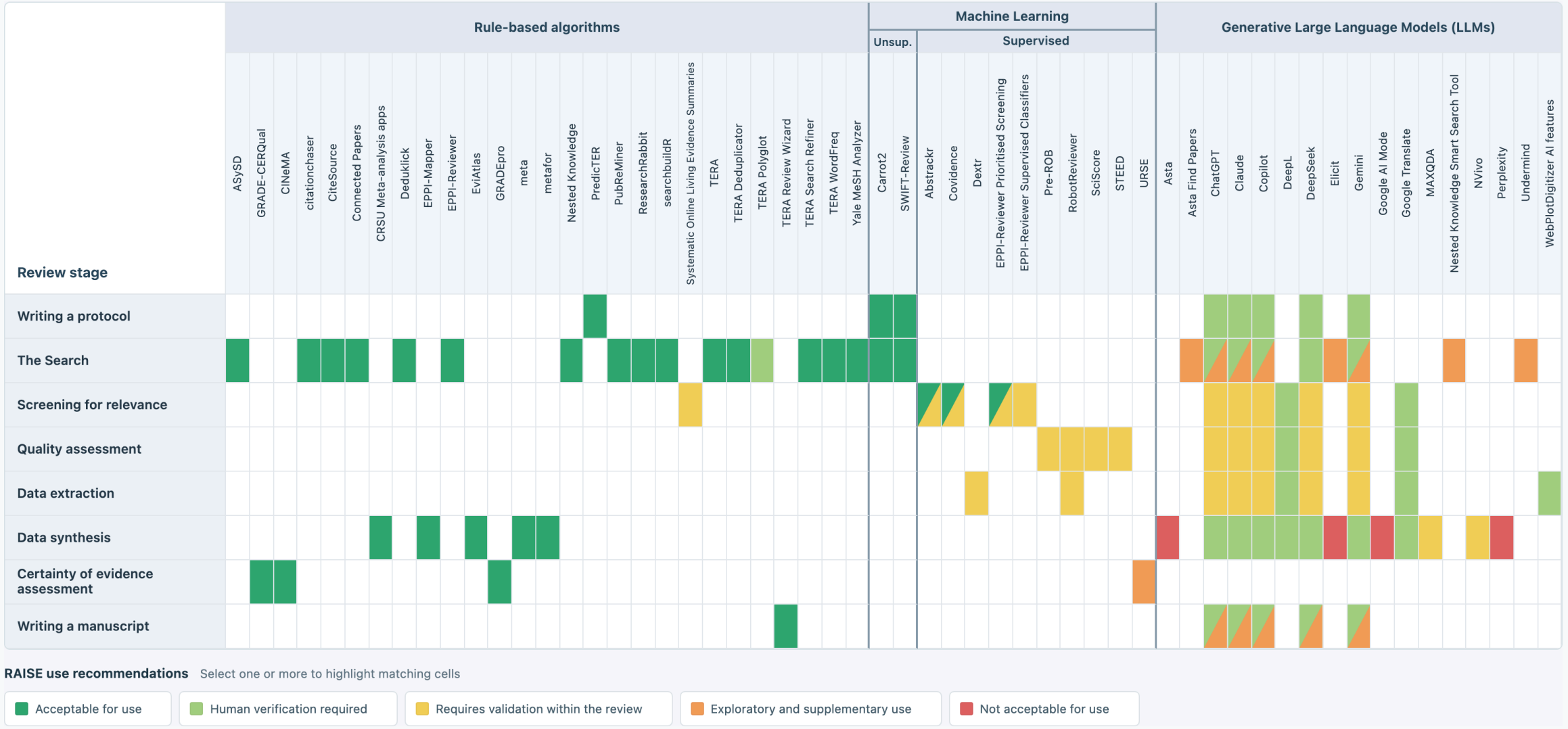
Recall: proportion of eligible records found

Applicable prior evidence may suffice. Follow any within-review requirement.

Suitable sampling or stopping methods can help, but assessment has a workload cost.

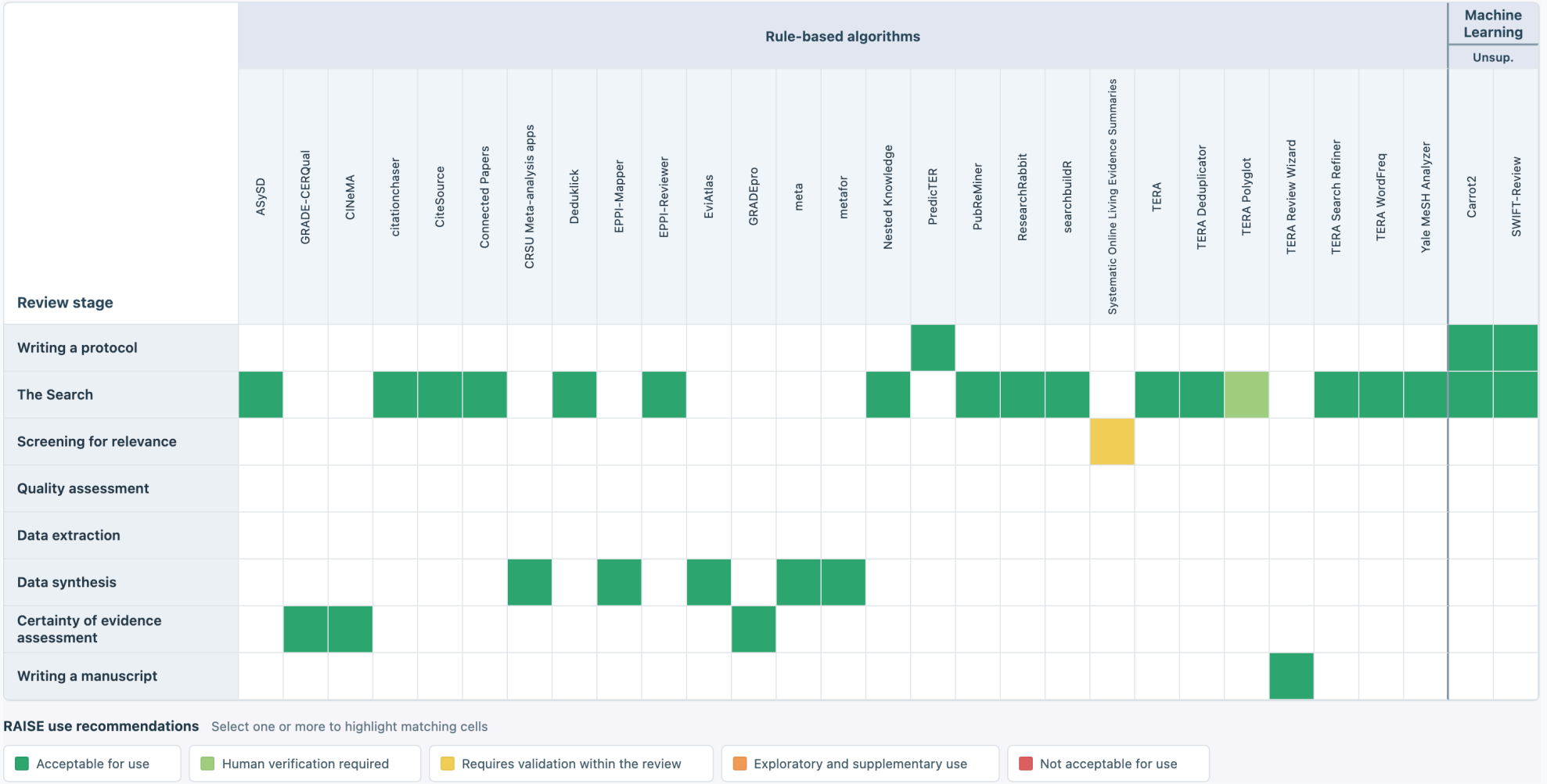
A workflow may need both.

AI tools across the review stages



Rule-based algorithms and unsupervised machine learning

Most of these mapped uses are acceptable



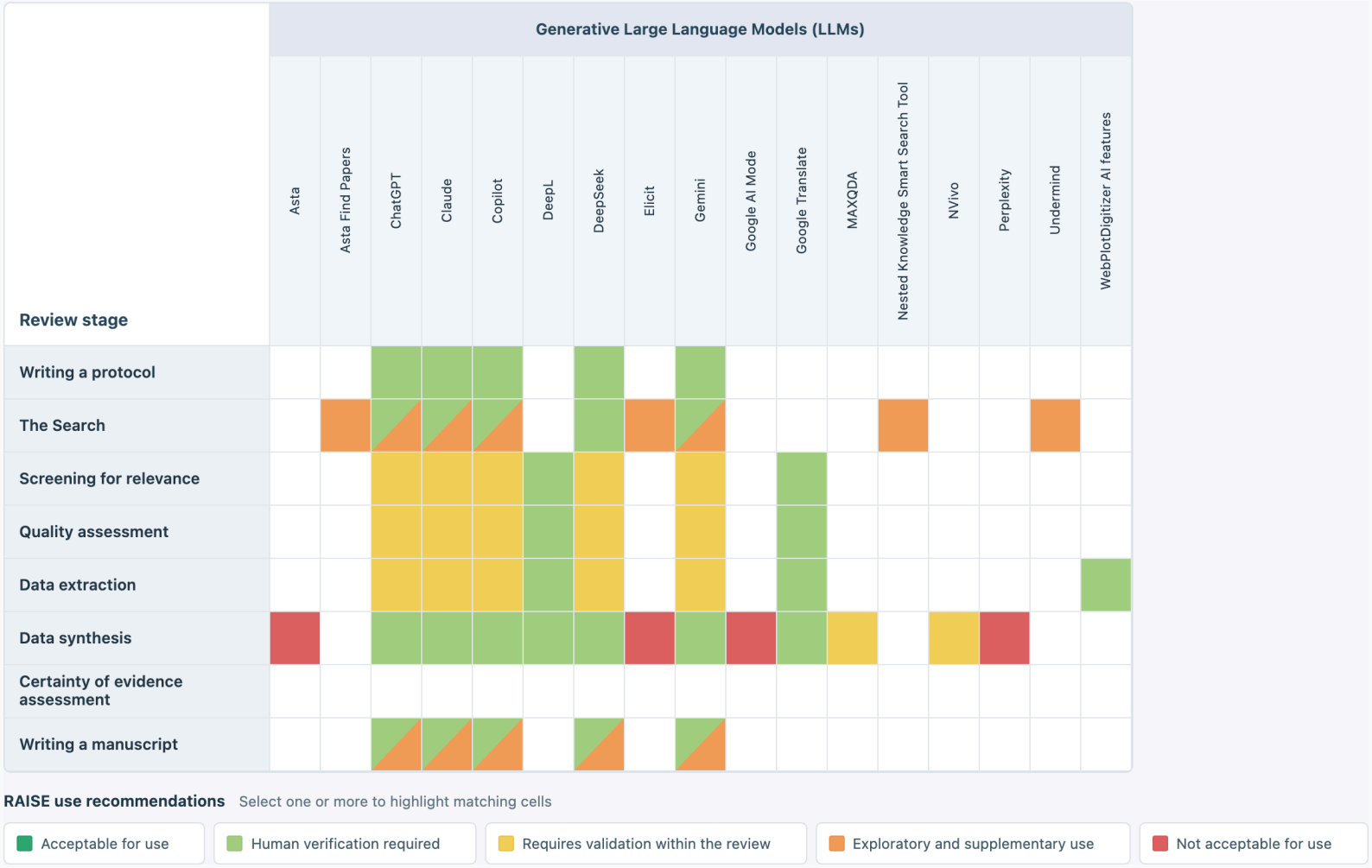
Supervised machine learning

Most mapped uses require validation within the review



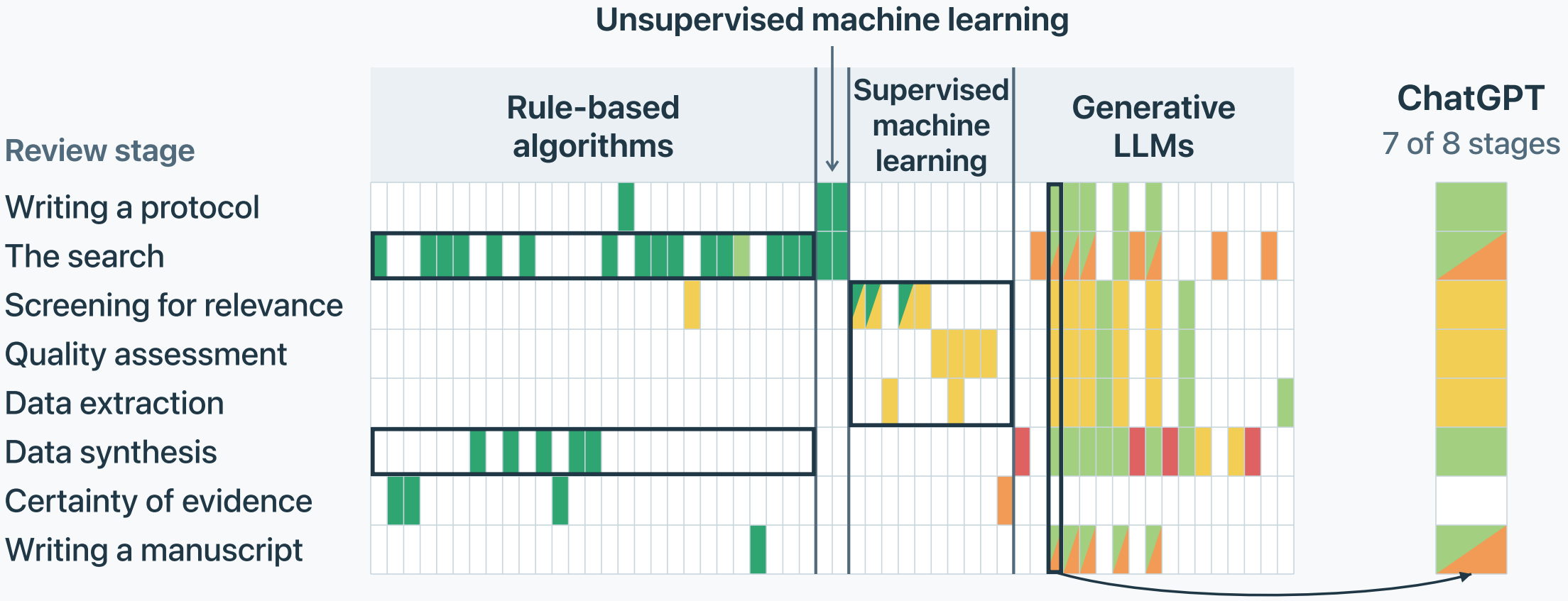
Generative large language models (LLMs)

The mapped uses carry conditions or restrictions



Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside



Rule-based
Search and synthesis

Supervised
Screening, quality assessment, extraction

ChatGPT
Mapped in 7 of 8 stages

Prioritised screening with and without early stopping

Stopping early changes reliance on the ranking

Abstrackr



Covidence



EPPI-Reviewer Prioritised Screening



Screen every record

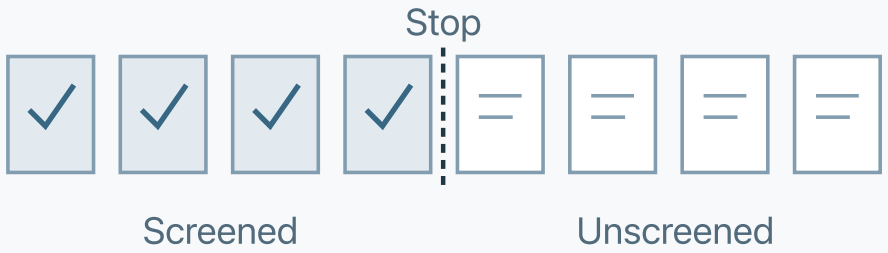


Every record is screened

Acceptable for use

A poor ranking does not leave records unseen.

Stop early



Requires validation within the review

Assess the screening and stopping procedure.

The ranking can be identical. What changes is how much we rely on it.

Two different uses during screening

Screening for relevance

Google Translate

Translation support

Human verification required

Check the translation against the original.

Google’s Gemini

Relevance judgements

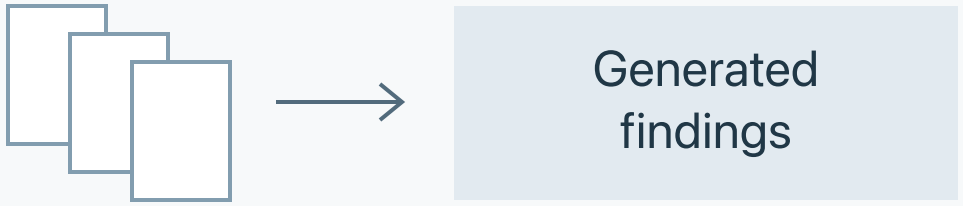
Requires validation within the review

Evaluate the screening procedure’s performance.

Selected screening cells from the map.

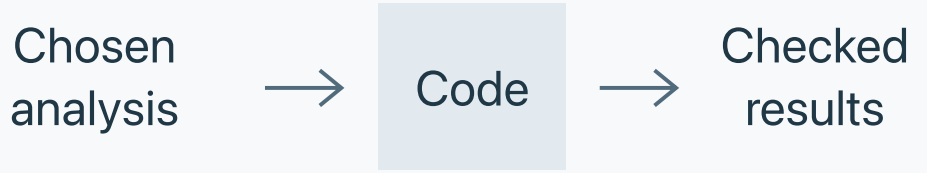
Synthesising findings differs from drafting analysis code

LLM-derived synthesis across studies



Not acceptable for use in a formal review

LLM-drafted code for a chosen analysis

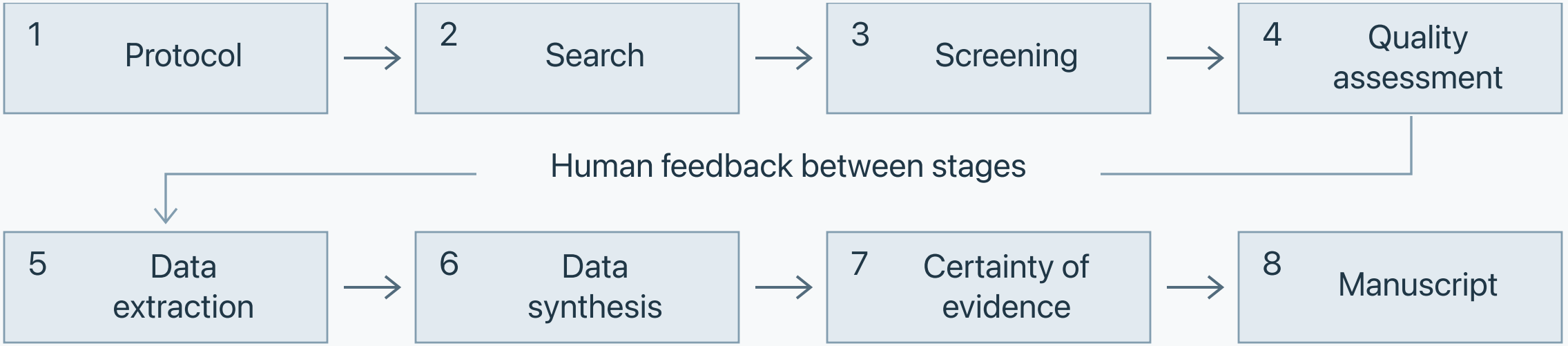


Human verification required

Qualitative coding or themes: validation and verification.
The boundary with delegated synthesis is not fully defined.

RAISE also restricts complete agentic reviews

Complete agentic review: not acceptable for use



Human feedback does not remove this restriction.

Individual tasks need separate assessment.

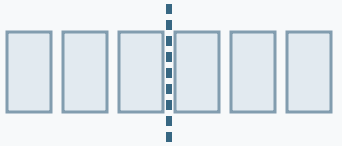
Is this tool suitable for my review?

Prioritised screening with early stopping

1

What is the purpose of the tool?

Assess ranking plus the stopping procedure.



2

Where have the training, testing and validation data come from?

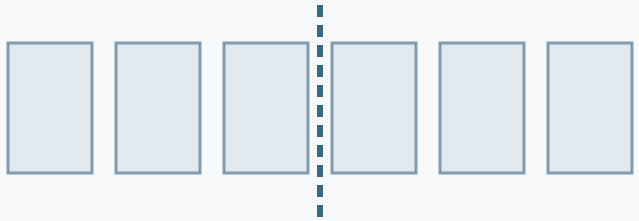
Review-team labels: quality and selection.
Method evaluations: relevance to this review.



Validation and performance

Prioritised screening with early stopping

3 Is the tool validated and performing sufficiently well for use?



Stopping procedure

Compare performance

Human-screened reference

Evidence for the stopping procedure

Set required performance in advance.
Establish whether it is met within this review.

An appropriate human-screened reference

Team, resources and documentation

Prioritised screening with early stopping

4 **Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?**

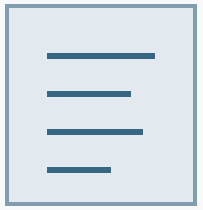
Skills, assessment time and a workable process

5 **Is there sufficient transparency around licenses and other available documentation?**

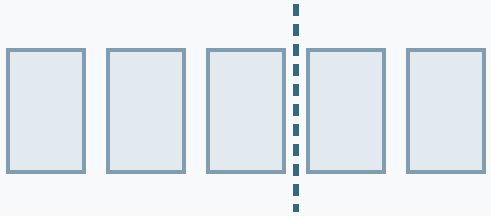
Instructions, access, data handling, version and settings

Missing stopping evidence must be resolved before reliance.

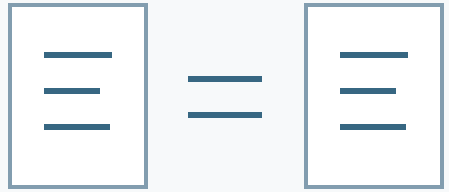
Choosing a suitable use of AI



Tool and task



Reliance on outputs



Evidence and checks

Map and presentation

lukebax.github.io/ai-review-tools/